

GAURAV TIWARI

DATA ENGINEER

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SUMMARY

Final-year B.Tech CSE student who builds end-to-end batch data pipelines in Python and SQL. Built a 6-stage orchestrated pipeline landing 100,000 rows into a 4-table PostgreSQL model with idempotent re-runs, and a Redis-coordinated distributed crawler with robots.txt-compliant per-domain rate limiting. Currently working with Kafka, Spark Streaming, Airflow, dbt and Snowflake. Seeking a full-time Data Engineer role, 2027 graduating batch.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C++

Data Engineering: ETL/ELT design, pipeline orchestration, batch and stream processing, data cleaning and validation, Apache Kafka, Spark Streaming, Airflow, dbt

Data Stores & Modeling: PostgreSQL, MongoDB, Redis, SQLite, Snowflake, Amazon Redshift, star schema and dimensional modeling, schema design

Python Data Stack: Pandas, NumPy, Scikit-Learn, XGBoost, Flask, Streamlit, BeautifulSoup, Matplotlib, Plotly

Cloud & Tooling: Docker, Docker Compose, Git, Linux, REST APIs, AWS (S3, Redshift), Supabase, Vercel

Foundations: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Distributed Systems

PROJECTS

Flight Delay Data Pipeline & Analytics Store | [GitHub](#) | [Live Demo](#)

Mar 2026

Python, Pandas, PostgreSQL, Scikit-Learn, XGBoost, Flask, Streamlit

- Built a 6-stage pipeline orchestrator — generate, clean, feature-engineer, load, train, evaluate — with `--skip-data` and `--skip-train` flags so a re-run reuses existing artifacts instead of regenerating 100,000 rows.
- Modelled a 4-table PostgreSQL layer separating raw, processed, prediction and run-metrics data, bulk-inserted each stage, and added automatic SQLite fallback so the pipeline runs end to end with no database server configured.
- Generated a 100,000-row synthetic dataset combining flight schedule, weather and airport-congestion signals, then applied median/mode imputation, IQR outlier clipping and duplicate removal as the cleaning stage.
- Engineered 18 features including cyclical sin/cos encoding of departure hour, a weighted weather-severity composite and route- and carrier-level historical mean delays, then selected XGBoost over Logistic Regression and Random Forest on 5-fold cross-validation at ROC-AUC 0.710, served behind a 5-endpoint Flask REST API and a Streamlit dashboard.

Distributed Web Crawler | [GitHub](#)

May 2026

Python, Redis, MongoDB, BeautifulSoup, Docker Compose, Streamlit

- Coordinated 5 worker threads through a Redis sorted-set frontier that ranks seed URLs above links discovered mid-crawl, plus a Redis visited-set index marked at dequeue time so concurrent workers never crawl the same page twice.
- Enforced politeness with a per-domain robots.txt parser cached on first fetch and a 1.0-second minimum gap between requests to the same domain, capping throughput by design rather than by worker count.
- Persisted parsed titles, meta descriptions, readable text and links to MongoDB, shipping Redis and MongoDB as Docker Compose services with live queue depth and pages-crawled on a Streamlit dashboard.

Real-Time Retail Data Pipeline

Mar 2026

Apache Kafka, Spark Structured Streaming, Apache Airflow, Amazon Redshift, S3, Python

- Ingested retail sales transactions through an Apache Kafka topic into Spark Structured Streaming, processing each micro-batch on arrival rather than on a fixed batch schedule.
- Landed intermediate output in Amazon S3 ahead of the Redshift load so a failed load can be replayed from staged files instead of re-reading the stream.
- Orchestrated the ingest and load stages as Apache Airflow DAGs with task-level retries, so a transient sink failure re-runs only the failed task rather than the whole pipeline.

End-to-End Data Warehouse for Flight Analytics

Mar 2026

Snowflake, dbt, Python, SQL, Apache Airflow, Tableau

- Modelled a Snowflake warehouse on a star schema over flight-operations and passenger sources, with dimension tables around a fact table at one-row-per-flight-leg grain.
- Built ELT pipelines in Python and dbt transforming raw loads into layered staging-to-marts models wired by dbt ref() dependencies, scheduled with Airflow and surfaced in a Tableau dashboard.

InvestIQ — Multi-Stage Research Pipeline | [GitHub](#) | [Live Demo](#)

Jun 2026

Next.js, TypeScript, LangGraph.js, Google Gemini, Supabase Postgres, Server-Sent Events

- Ingested live company data from the Tavily search API and Yahoo Finance into a 5-node LangGraph state graph, with a fan-out stage issuing 5 analysis calls concurrently off a single shared research fetch, and a bounded reviewer critique cycle that terminates instead of looping.

CERTIFICATIONS

• **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — Oracle

Aug 2025

• **Cloud Computing** — NPTEL

Nov 2025

• **CodeQuest DSA Summer Bootcamp** — 100+ problems across arrays, graphs and DP

Jun 2025

EDUCATION

Lovely Professional University, Phagwara, Punjab

Aug 2023 - May 2027

B.Tech, Computer Science and Engineering | CGPA 7.26

Lions School, Mirzapur, Uttar Pradesh

Mar 2019 - May 2022

Intermediate (PCM), 74.6%